

ALPHA-GPT 3.0: A MULTI-AGENT FRAMEWORK FOR QUANTITATIVE INVESTMENTS

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ABSTRACT

In quantitative investment, much of the research effort goes into discovering alphas, which are symbolic formulas that capture predictive signals in market data. There are two main approaches to finding these: domain experts manually craft them based on financial intuition, or algorithms like symbolic regression over massive expression spaces. Both have clear drawbacks. Manual alpha mining is slow, expensive, and hard to scale, while algorithmic search burns through compute and produces formulas that are difficult to interpret. Alpha-GPT showed that large language models can bridge this gap by translating trading ideas written in natural language into valid formulaic expressions, and then feeding those into an algorithmic search pipeline. We build on this work by implementing a multi-agent system where separate LLM agents each own a stage of the pipeline: (i) instead of relying on a single model to convert trading ideas into seed formulas, we instantiate multiple LLM agents debate framework that collaboratively reason over each input idea, (ii) another agent configures and manages the genetic programming search, and (iii) third agent analyzes results and explains them back in plain language. We test our system on U.S. equity data and evaluate using standard quant metrics like out-of-sample information coefficient and Sharpe ratio. Beyond reproducing the core Alpha-GPT pipeline, we study how debate-based generation affects expression and how different prompt engineering strategies affect the validity and quality of generated expressions, compare multi-agent orchestration against a single-agent baseline, and characterize common failure modes in LLM-generated alphas.

1 INTRODUCTION

Designing quantitative equity factors is an inherently iterative process. A trading idea expressed in natural language is rarely precise enough to be translated directly into a usable formula: the underlying economic mechanism may be ambiguous, the role of the signal may be unclear, and normalization often remain unresolved. As a result, much of alpha research effort is spent not on backtesting itself, but on turning informal investment theses into valid and testable factor expressions.

Large language models make this translation problem newly tractable. Finance-oriented models such as BloombergGPT and FinGPT show that LLMs can encode substantial domain knowledge and perform competitively on financial language tasks, suggesting that they may be useful for assisting quantitative research (19; 20). Prior empirical work also indicates that LLM-generated judgments can contain informative but unstable financial signal content (11). In the specific setting of alpha mining, Alpha-GPT introduces a human-AI interactive framework that translates natural-language trading ideas into formulaic alpha expressions, while Alpha-GPT 2.0 extends this line of work toward a broader human-in-the-loop quantitative research pipeline (15; 23). Together, these works

make LLM-based alpha research a plausible direction, but they also leave open the question of how to make generation more reliable.

In practice, direct one-shot generation is often brittle. A model may produce a formula that no longer matches the intended economic hypothesis, or it may fail at the implementation level through invalid operators, unavailable fields, or parser-unfriendly syntax. These failures are especially costly in alpha research because unusable expressions are filtered out before backtesting, reducing effective research throughput. More broadly, the challenge is not only to generate formulas, but to preserve traceability from the original idea to the final symbolic expression.

To address this issue, we build on Alpha-GPT and introduce a multi-agent debate framework for factor generation. Our key idea is to separate conceptual interpretation from symbolic expression writing, rather than asking a single model to perform both tasks in one pass. In Stage 1, which we call Idea Debate, multiple agents independently interpret the same natural-language thesis, critique one another’s reasoning, and converge to a small set of structured hypothesis specifications. In Stage 2, Formula Debate, agents translate these specifications into candidate formulas, review one another for faithfulness and implementability, and revise them before final selection. This two-stage design improves traceability, interpretability, and debugging relative to direct one-shot generation.

We study whether structured multi-agent deliberation can improve LLM-based factor generation along three practically important dimensions: validity, diversity, and downstream predictive usefulness. To answer this question, we build a unified evaluation framework in which debate-based generation is compared with simpler alternatives, including single-agent prompting and heuristic symbolic-search baselines, under the same data pipeline and out-of-sample testing protocol. This setup allows us to isolate the effect of deliberative generation itself, rather than confounding it with differences in data handling or evaluation design.

2 RELATED WORK

Financial NLP has evolved from task-specific encoders such as FinBERT (1) to domain-adapted large language models. BloombergGPT demonstrates that large-scale pretraining on mixed financial and general corpora can materially improve finance-task performance without sacrificing broad language capability (19). FinGPT provides an open-source route for financial instruction tuning and continual adaptation (20). Beyond model-building, return-prediction studies using ChatGPT-style models suggest that LLM semantic judgments on news can contain incremental forecasting information (11). This literature establishes that LLMs can be useful in financial decision workflows, but most work focuses on prediction, sentiment, or question-answering tasks.

Most directly related to our work, Alpha-GPT proposes a human-AI interactive alpha mining paradigm in which LLMs help translate researchers’ ideas into formulaic alpha expressions (15). Alpha-GPT 2.0 expands this direction into a broader human-in-the-loop quantitative investment framework that covers additional modeling and analysis stages beyond initial alpha proposal (23). These papers establish the relevance of LLMs for alpha research, but they primarily emphasize interaction and pipeline integration. Our work instead focuses on improving generation reliability through structured multi-agent deliberation, with explicit separation between hypothesis interpretation and formula construction.

Another relevant line of work studies prompting and inference strategies that improve reasoning reliability. Chain-of-Thought prompting and self-consistency show that explicit multi-step reasoning and diversified sampling can improve performance on difficult reasoning tasks (17; 16). Tree-of-Thought and ReAct further emphasize deliberate search over intermediate reasoning states and the interaction between reasoning and action (21; 22). Iterative self-feedback approaches such as Self-Refine and Reflexion suggest that critique-and-revise loops can improve output quality without task-specific retraining (12; 14). These works motivate our staged critique process, in which agents revise both hypotheses and formulas across rounds instead of producing one-shot outputs.

Recent multi-agent systems ask whether distributing cognition across interacting agents yields more robust outcomes than single-model prompting. CAMEL introduces role-playing agent societies for cooperative problem solving (10); AutoGen provides abstractions for message-passing and tool-using agent workflows (18); ChatDev and MetaGPT demonstrate structured role decomposition in complex software tasks (13; 7); and AgentVerse studies coordination behavior in larger agent popu-

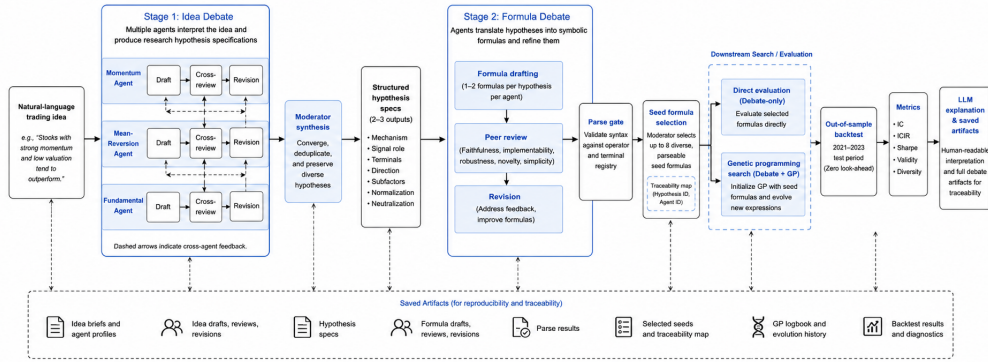


Figure 1: Overall workflow of Alpha-GPT 3.0. The system converts a natural-language trading idea into testable alpha signals through a two-stage multi-agent debate framework, parse-gated selection, optional genetic programming search, and out-of-sample evaluation.

Figure 1: The general pipeline of alpha-gpt3.0

lations (3). Although these frameworks are largely domain-agnostic, they provide design principles that are useful in quantitative finance, including heterogeneous priors, explicit communication protocols, and moderator-style coordination.

Debate-based methods specifically frame disagreement as a mechanism for quality control. Multi-agent debate has been shown to improve factuality and reasoning relative to isolated generation (4). Related work such as ChatEval uses debate among agents to improve evaluation reliability, reinforcing the idea that adversarial critique can surface hidden errors (2). This perspective is particularly relevant for alpha research, where invalid assumptions, weak proxies, and non-implementable operators are often subtle and may not be caught in a single generation pass. Our two-stage framework adapts this idea to the factor-generation setting by separating debate over what should be tested from debate over how it should be encoded.

Formulaic alpha research emphasizes interpretable symbolic factors as reusable research objects (8). Heuristic symbolic-search methods, including genetic programming (9) and its practical implementations such as DEAP (5), remain useful baselines for exploring factor spaces without language-based reasoning. In our study, these methods are treated as comparison baselines rather than as the core contribution, since our emphasis is on whether deliberative LLM generation improves the validity and usefulness of generated factors.

3 METHOD

3.1 FRAMEWORK OVERVIEW

Our system extends the Alpha-GPT pipeline by introducing a structured two-stage multi-agent debate framework that sits between natural-language idea input and genetic programming search. Rather than asking a single model to translate an informal trading thesis directly into a formula, we decompose the generation process into two sequential debates: the first resolves how to interpret the original hypothesis as a structured research specification, and the second determines how to express that specification as a concrete symbolic formula. This decomposition is motivated by the observation that failures in direct one-shot generation often arise from the conflation of two distinct reasoning tasks. A model simultaneously reasoning about economic mechanism, data availability, and operator syntax tends to produce outputs that are locally plausible but globally inconsistent—for example, a formula that implements a coherent strategy in principle but uses terminals that are unavailable in the data, or one that inverts the intended direction of prediction. By staging the process, each class of failure becomes visible and addressable independently.

Figure 1 summarizes the full Alpha-GPT 3.0 workflow from natural-language idea input to formula selection, optional genetic programming, and out-of-sample evaluation.

3.2 AGENT DESIGN

Three specialized agents participate in both stages of the debate. Each agent is initialized with a distinct investment prior encoded in its system prompt. The momentum agent reasons primarily from trend persistence, price strength, breakout dynamics, and volume confirmation patterns, biasing its proposals toward signals that exploit continuation in cross-sectional return rank. The mean-reversion agent emphasizes overshooting, contrarian positioning, temporary dislocations, and volatility-induced corrections, and is predisposed to propose signals that exploit short-run reversals or spread normalization. The fundamental agent takes a valuation and quality perspective, grounding its reasoning in earnings yield, balance sheet strength, profitability ratios, and the slower pace at which fundamental information is incorporated into prices. Despite these stylistic differences, each agent is a complete researcher expected to reason across all dimensions of a factor proposal, including signal role, data field selection, directionality, normalization, neutralization, and implementability under the available operator set. The use of three agents with structurally distinct priors is designed to surface a wider range of candidate interpretations than single-agent prompting, while keeping the total number of LLM calls tractable.

3.3 STAGE 1: IDEA DEBATE

Given a natural-language trading idea, Stage 1 produces two to three structured research hypothesis specifications through a three-phase process: drafting, cross-review, and revision with moderator synthesis.

In the drafting phase, each agent independently generates an idea proposal that includes an explicit characterization of the economic mechanism, the intended signal role (cross-sectional or time-series), the specific terminal variables required, the direction of the predicted relationship, any sub-factors or conditioning filters, and a plan for normalization and neutralization. Agents are given hard constraints specifying only the terminals available in the data panel and are prohibited from embedding numerical formulas at this stage. Each proposal is assigned a unique identifier to support traceability throughout the subsequent debate.

In the cross-review phase, each agent evaluates the proposals produced by the other two agents. Reviews are structured across eight scoring dimensions: mechanism quality, signal-type clarity, payoff clarity, directionality clarity, subfactor quality, filter logic, normalization soundness, and implementability. Each dimension is scored on a zero-to-five integer scale. Reviewers also render a categorical decision—accept, accept with revision, or reject—accompanied by free-form comments justifying the scores and flagging specific weaknesses.

In the revision phase, each agent receives the two reviews of its own draft and produces a revised proposal, explicitly recording which feedback was accepted, which was rejected, and summarizing the changes made. This explicit tracking is important for interpretability: it allows downstream analysis to identify which agent’s critique most frequently led to substantive revision, and which types of weaknesses were most commonly raised. The three revised proposals are then passed to a moderator agent, which synthesizes them into two to three final research hypothesis specifications. The moderator is instructed to converge the debate rather than to introduce new ideas: it preserves meaningful distinctions across the revised proposals, removes near-duplicates arising from similar priors, and annotates each output hypothesis with the identifiers of the source agents whose reasoning it reflects.

3.4 STAGE 2: FORMULA DEBATE

Stage 2 takes the research hypothesis specifications produced in Stage 1 and generates a set of concrete symbolic formulas through an analogous three-phase debate.

In the drafting phase, each agent produces one to two formula candidates per hypothesis. All formulas are written using a shared operator set comprising more than 35 curried operators and 34 terminal variables corresponding to the available data fields. Each candidate is tagged with a formula role—main alpha, directional alpha, filter, or composite—along with a rationale explaining how the expression encodes the corresponding hypothesis and why the chosen operators and terminals are appropriate.

In the cross-review phase, each agent evaluates the formula candidates produced by the other two agents. Reviews are structured across five dimensions: faithfulness to the originating hypothesis, implementability under the operator and terminal constraints, robustness to noise and outliers, novelty relative to standard factors, and syntactic simplicity. Categorical decisions and free-form comments are provided as in Stage 1, with particular attention to whether the formula correctly reflects the intended signal direction and whether it introduces subtle look-ahead through the choice of aggregation windows.

In the revision and selection phase, each agent revises its own formula candidates in response to the received reviews. Before moderator selection, all revised formulas undergo a parse-gating step in which each expression is evaluated against the live operator and terminal registry; only expressions that parse successfully are eligible for final selection. This step provides a hard, model-independent filter against syntactically invalid or operator-misusing candidates. The moderator then selects a final seed set of up to eight formulas from the parseable candidates, with an explicit preference for diversity across hypotheses and signal roles. The selection output includes a traceability map that links each selected formula back to its originating hypothesis and agent, enabling downstream failure analysis to be traced to specific stages and reasoning chains within the pipeline.

3.5 DOWNSTREAM GENETIC PROGRAMMING

The seed formulas produced by the debate pipeline initialize a genetic programming search. Beginning from semantically grounded seeds rather than random expression trees is intended to concentrate the search in regions of formula space that are already consistent with the original trading thesis, reducing the expected number of fitness evaluations required to reach formulas with positive out-of-sample predictive signal. The GP search uses tournament selection, standard subtree crossover and point mutation over the same shared operator and terminal set, and is evaluated using Spearman IC on the training split only. Factor quality is assessed on the held-out test period to prevent any leakage of test-period signal into the generation or selection process.

4 EXPERIMENT

4.1 DATASET

Our evaluation is conducted on a panel of 500 U.S. common equity securities drawn from the CRSP/Compustat merged database, covering the period from January 2010 through December 2023. The universe is constructed by stratifying eligible equities into market-capitalization quintiles and sampling 100 securities per quintile at the start of each calendar year, ensuring that the panel preserves cross-sectional diversity across firm size and is not dominated by large-cap names. Securities that are delisted or cease trading within a calendar year are retained up to their last active trading day and are not replaced mid-year, limiting survivorship bias to the annual rebalancing step.

Market variables are sourced from CRSP and include daily open, high, low, and closing prices, unadjusted and adjusted return series, trading volume, and total shares outstanding. Factor-relevant derived quantities include dollar volume, share turnover, and price momentum at multiple look-back horizons. Accounting variables are sourced from Compustat quarterly filings and include the book-to-market ratio, return on equity (ROE), return on assets (ROA), price-to-earnings ratio (PE), price-to-book ratio (PTB), gross margin, operating margin, net margin, total debt-to-equity, interest coverage ratio, and the current ratio as a measure of short-term liquidity. Because accounting data are reported at quarterly frequency, all fundamental variables are forward-filled to daily frequency beginning at each quarter-end reporting date, reflecting the information set available to an investor on each trading day and following standard practice in the empirical factor literature (6). No future accounting observations are introduced into any historical trading date.

Raw variables are preprocessed cross-sectionally before factor evaluation. Extreme values are winsorized at the 1st and 99th percentiles on each trading day to reduce the influence of outliers and data errors. All cross-sectional distributions are then standardized to zero mean and unit variance. Factor expressions are applied to these preprocessed variables, and the resulting factor scores are rank-transformed cross-sectionally before portfolio assignment to further reduce sensitivity to distributional assumptions.

The full data panel is partitioned strictly by calendar time into three non-overlapping periods. The training period spans January 2010 through December 2017 and is used for genetic programming fitness evaluation and for any hyperparameter choices made during the generation process. The validation period spans January 2018 through December 2020 and is used for model selection among generation strategies. The held-out test period spans January 2021 through December 2023 and is used exclusively for final evaluation; no design decision in the generation pipeline or baseline configuration has access to test-period outcomes.

4.2 EVALUATION PROTOCOL

We compare factor-generation methods under a unified pipeline. For each trading idea, candidate formulas are generated by (i) our two-stage multi-agent debate framework, (ii) a single-agent zero-shot baseline, and (iii) a random-initialized GP baseline. All candidates are parsed, validated under the same operator and terminal constraints, and evaluated on the held-out test period (2021–2023), so that differences in outcome can be attributed to the generation method itself. All methods use daily long-short quintile portfolios with equal-weighted positions in the top and bottom signal quintiles.

4.3 EVALUATION METRICS

We evaluate generated factors along three complementary dimensions: raw predictive signal quality, signal stability, and portfolio-level performance.

Information Coefficient (IC). The information coefficient measures the cross-sectional predictive accuracy of a factor on a single trading day. Let $\alpha_t \in \mathbb{R}^N$ denote the vector of factor scores for N stocks on day t , and let $\mathbf{r}_{t+1} \in \mathbb{R}^N$ denote the corresponding vector of next-day returns. The daily IC is defined as the Spearman rank correlation between α_t and $\mathbf{r}_{t+1}$:

$$\text{IC}_t = \rho_s(\alpha_t, \mathbf{r}_{t+1}) = 1 - \frac{6 \sum_{i=1}^N d_i^2}{N(N^2 - 1)}, \quad (1)$$

where $d_i = \text{rank}(\alpha_{t,i}) - \text{rank}(r_{t+1,i})$ is the rank difference for stock i . Rank correlation is used rather than Pearson correlation to reduce sensitivity to return outliers and to avoid distortion from a small number of extreme observations (6). The summary statistic reported is the time-series mean $\overline{\text{IC}} = T^{-1} \sum_{t=1}^T \text{IC}_t$ over the evaluation period. Positive IC indicates that higher factor scores tend to predict higher subsequent returns in the intended direction; values consistently above 0.02 are generally regarded as economically meaningful in practice, as they correspond to a systematic cross-sectional signal above the level of short-run noise.

IC Information Ratio (ICIR). Mean IC alone does not distinguish a factor that produces a steady, consistent predictive signal from one that delivers occasional large ICs interspersed with many negative or near-zero values. The ICIR corrects for this by normalizing mean IC by its temporal variability:

$$\text{ICIR} = \frac{\overline{\text{IC}}}{\sigma_{\text{IC}}}, \quad (2)$$

where σ_{IC} is the sample standard deviation of the daily IC time series over the evaluation period. ICIR is therefore analogous to a signal-to-noise ratio for factor predictability: a high ICIR indicates that the factor produces reliably positive predictive signal across different market regimes rather than concentrating its predictive power in a small number of favorable periods. In practice, factors with $\text{ICIR} > 0.5$ are generally considered stable enough to be deployable, as the implied variability of the signal is low relative to its mean level.

Sharpe Ratio. Portfolio-level performance is evaluated using the annualized Sharpe ratio of a daily long-short quintile portfolio. On each day t , stocks are ranked cross-sectionally by their factor scores, and equal-weighted long positions are taken in the top quintile while equal-weighted short positions are taken in the bottom quintile. Let r_t^{LS} denote the return of this long-short portfolio on

day t . The annualized Sharpe ratio is:

$$SR = \sqrt{252} \cdot \frac{\mathbb{E}[r_t^{LS}]}{\sqrt{\text{Var}(r_t^{LS})}}, \quad (3)$$

where the factor $\sqrt{252}$ converts from daily to annualized units, reflecting the number of trading days per year. The risk-free rate is set to zero, which is standard for long-short equity factor evaluation where the portfolio is approximately dollar-neutral and the financing cost of the short leg is netted against the long leg implicitly. The Sharpe ratio integrates both the level and consistency of factor returns into a single statistic, and serves as the primary portfolio-level criterion for comparing generation strategies.

Formula Validity Rate. In addition to prediction quality, we report the validity rate of generated formulas, defined as the proportion of candidate expressions that parse successfully against the operator and terminal registry prior to any downstream evaluation:

$$\text{Validity} = \frac{|\{e_i : \text{parse}(e_i) = \text{success}\}|}{|\{e_i\}|}, \quad (4)$$

where $\{e_i\}$ denotes the full set of candidate formulas produced by a given generation method. This metric directly measures the implementation quality of LLM generation: a high validity rate indicates that the model correctly respects the operator grammar and terminal constraints, whereas a low rate signals that a substantial fraction of generated candidates are unusable without manual correction. Validity rate is particularly relevant for the comparison between multi-agent debate and single-agent zero-shot generation, as the debate process incorporates structured implementability review at both stages and parse-gating before moderator selection, all of which are absent from the single-agent baseline.

Formula Diversity. Validity alone does not guarantee that a generation method explores a useful range of candidate signals. We therefore also track formula diversity using operator coverage, terminal coverage, and duplicate-expression rate. Operator coverage measures the fraction of available operators used by at least one generated candidate, while terminal coverage measures the fraction of available data fields used by at least one generated candidate. The duplicate-expression rate measures the proportion of generated formulas that are exact repeats after expression normalization. These metrics are especially relevant for comparing debate-based generation against single-agent generation, since heterogeneous agent priors are intended to broaden the set of tested hypotheses rather than merely improve syntax.

Traceability. For debate-based methods, we additionally record whether each selected formula can be traced back to a structured hypothesis and source agent. A formula is considered traceable if the saved debate artifact links the final expression to a hypothesis identifier produced in Stage 1 and a formula proposal produced in Stage 2. This metric captures a practical advantage of the debate pipeline: when an alpha fails or behaves unexpectedly, the researcher can inspect whether the error originated from hypothesis interpretation, formula construction, peer review, or moderator selection.

4.4 SINGLE-AGENT BASELINE (ZERO-SHOT)

A single LLM generates candidate factor formulas from a natural-language trading idea in one pass, without debate, critique, or revision. The outputs are parsed, validated, and evaluated under the same downstream pipeline as our method. This baseline isolates the contribution of multi-agent deliberation by holding the generation model and data pipeline constant while removing the debate structure.

4.5 RANDOM GP BASELINE (HEURISTIC)

Candidate factors are generated by genetic programming initialized from fully random expression trees, with no LLM involvement. The same operator set, terminal set, validity constraints, and evaluation pipeline are used as for all other methods. This baseline isolates the value of language-guided generation relative to unguided symbolic search, and establishes the lower bound of what structured reasoning over trading ideas is expected to improve upon.

4.6 IMPLEMENTATION DETAILS

All LLM calls use DeepSeek-V3.2 via the OpenRouter API with temperature 0.4. The single-agent baseline generates up to 8 candidate formulas per call. For GP baselines, we use a population of 50, 10 generations, tournament selection (size 3), crossover probability 0.7, mutation probability 0.2, and maximum tree depth 6. Fitness is mean Spearman IC on the training set. Each configuration is run 3 times with fixed random seeds for reproducibility. All formulas are evaluated on the same 34-terminal operator set comprising 35+ curried operators.

4.7 QUALITATIVE CASE STUDY

Table 1 shows an example of how the system converts a natural-language trading idea into a structured formula. The example illustrates the intended traceability of the pipeline: the final expression can be connected back to a hypothesis about improving profitability and liquidity stability, rather than appearing as an opaque symbolic artifact.

Table 1: Example generated alpha and interpretation.

Field	Example
Trading idea	Stocks with strong recent momentum and low valuation tend to outperform.
Hypothesis	Firms with improving profitability and stable liquidity may be rewarded by the market.
Generated formula	<code>ts_min_20 (add (add (ts_max_5 (ts_delta_20 (roe)), ts_std_5 (curr_ratio)), ts_max_5 (add (ts_delta_20 (roe), roe))))</code>
Interpretation	The formula combines changes in return on equity with liquidity stability. It favors firms whose profitability is improving while avoiding firms with unstable short-term liquidity conditions.
Observed risk	The factor has positive IC but large drawdowns, suggesting regime sensitivity and the need for portfolio-level risk controls.

4.8 RESULTS

Table 2 reports median metrics across the top 5 alphas per run, aggregated (mean $\pm$ std) over 3 runs.

Table 2: Baseline comparison on held-out test period (2021–2023).

Mode	IC	ICIR	Sharpe
Random GP	0.0285 ± 0.0154	0.1822 ± 0.0860	0.45 ± 0.30
Single-Agent	-0.0030 ± 0.0029	-0.0186 ± 0.0177	-0.45 ± 0.49
Single-Agent + GP	-0.0068 ± 0.0038	-0.0444 ± 0.0251	-0.56 ± 0.30

Figure 2 plots the cumulative out-of-sample performance of the top generated alpha signals from 2021 to 2023, together with the value-weighted market benchmark. The best-performing alpha produces a substantially stronger cumulative return than the market benchmark, reaching roughly 1.75 by the end of the test period compared with about 1.25 for the benchmark. The other generated alphas deliver more modest gains, indicating that the pipeline can identify a small number of highly effective candidates even when many formulas remain close to market-level performance. The figure also shows that the strongest alpha experiences nontrivial drawdowns, especially around late 2022, suggesting that predictive strength should be evaluated together with stability and risk metrics rather than cumulative return alone.

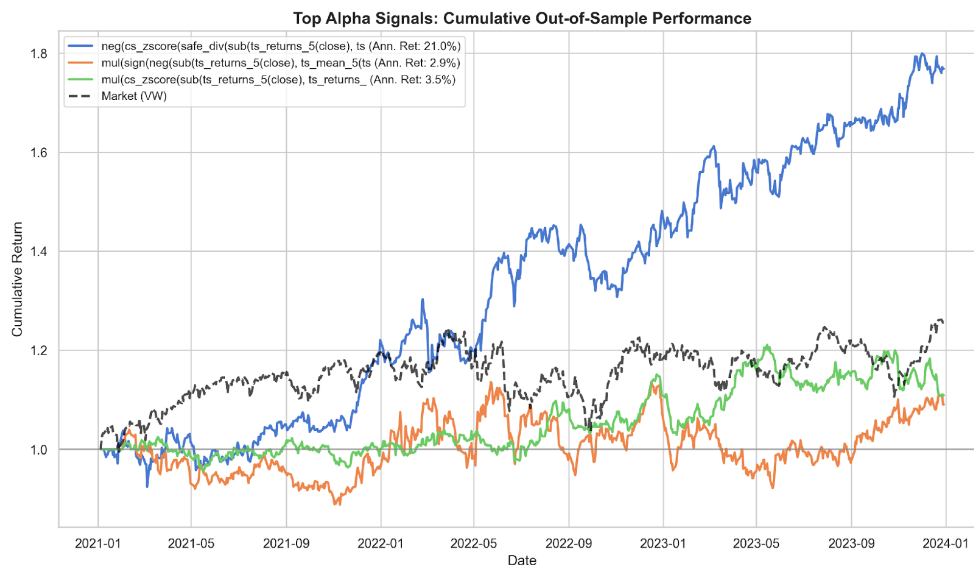


Figure 2: Cumulative out-of-sample performance of the top generated alpha signals over the 2021–2023 test period. Each colored line corresponds to a selected alpha formula, while the dashed black line reports the value-weighted market benchmark.

5 CONCLUSION

We introduce Alpha-GPT 3.0, a two-stage multi-agent debate framework for LLM-guided quantitative alpha generation. By separating hypothesis interpretation from symbolic formula construction, the system makes generated alphas more inspectable and easier to debug than one-shot LLM outputs. The framework also introduces practical quality-control mechanisms, including peer review, revision, moderator selection, parse-gating, and traceability from final formulas back to source hypotheses. Our preliminary baseline results show that language guidance alone is not sufficient: single-agent generation can underperform random symbolic search when generated seeds are invalid, poorly aligned, or overly brittle. This motivates evaluating alpha-generation systems not only by downstream Sharpe or IC, but also by formula validity, diversity, and traceability. Future work will expand the number of trading ideas, evaluate more market regimes, incorporate transaction costs and risk neutralization, and study whether debate-generated seeds improve GP search efficiency across a broader universe of alpha hypotheses.

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I acknowledge the use of ChatGPT (<https://chat.openai.com/>) and Codex (<https://openai.com/codex/>) to clarify code behavior, answer conceptual scientific questions, suggest workflow-figure layouts, polish English grammar, and revise LaTeX formatting. I used the outputs as consulting support to improve presentation, clarify technical details, and refine the wording and formatting of this document. I have the ability to explain and even independently replicate the work done in this document if asked by an instructor.

CONTRIBUTION STATEMENT

Hao Wang contributed to generating and modifying the codebase and writing the report. Shuai Meng contributed to generating and modifying the codebase and writing the report. Ronald Liu contributed to generating and modifying the codebase, downloading the data, and preparing the presentation slides.

REFERENCES

- [1] Dogu Araci. Finbert: Financial sentiment analysis with pre-trained language models. *arXiv preprint arXiv:1908.10063*, 2019.
- [2] Chi-Min Chan, Weize Chen, Yusheng Su, Jianxuan Yu, Wei Xue, Shanghang Zhang, Jie Fu, and Zhiyuan Liu. Chateval: Towards better llm-based evaluators through multi-agent debate. In *International Conference on Learning Representations (ICLR)*, 2024.
- [3] Weize Chen, Yusheng Su, Jingwei Zuo, Cheng Yang, Chenfei Yuan, Chi-Min Chan, Heyang Yu, Yaxi Lu, Yi-Hsin Hung, Chen Qian, Yujia Qin, Xin Cong, Ruobing Xie, Zhiyuan Liu, Maosong Sun, and Jie Zhou. Agentverse: Facilitating multi-agent collaboration and exploring emergent behaviors. In *International Conference on Learning Representations (ICLR)*, 2023.
- [4] Yilun Du, Shuang Li, Antonio Torralba, Joshua B. Tenenbaum, and Igor Mordatch. Improving factuality and reasoning in language models through multiagent debate. *arXiv preprint arXiv:2305.14325*, 2023.
- [5] Felix-Antoine Fortin, François-Michel De Rainville, Marc-André Gardner, Marc Parizeau, and Christian Gagné. Deap: Evolutionary algorithms made easy. *Journal of Machine Learning Research*, 13:2171–2175, 2012.
- [6] Shihao Gu, Bryan Kelly, and Dacheng Xiu. Empirical asset pricing via machine learning. *Review of Financial Studies*, 33(5):2223–2273, 2020.
- [7] Sirui Hong, Mingchen Zhuge, Jiaqi Chen, Xiwu Zheng, Yuheng Cheng, Ceyao Zhang, Jinlin Wang, Zili Wang, Steven Ka Shing Yau, Zijuan Lin, Liyang Zhou, Chenyu Ran, Lingfeng Xiao, Chenglin Wu, and Jürgen Schmidhuber. Metagpt: Meta programming for a multi-agent collaborative framework. *arXiv preprint arXiv:2308.00352*, 2023.
- [8] Zura Kakushadze. 101 formulaic alphas. *arXiv preprint arXiv:1601.00991*, 2016.
- [9] John R. Koza. *Genetic Programming: On the Programming of Computers by Means of Natural Selection*. MIT Press, 1992.
- [10] Guohao Li, Hasan Abed Al Kader Hammoud, Hani Itani, Dmitrii Khizbullin, and Bernard Ghanem. Camel: Communicative agents for “mind” exploration of large language model society. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*, 2023.
- [11] Alejandro López-Lira and Yuehua Tang. Can chatgpt forecast stock price movements? return predictability and large language models. *arXiv preprint arXiv:2304.07619*, 2023.
- [12] Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wier-effe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, Shashank Gupta, Bodhisattwa Prasad Majumder, Katherine Hermann, Sean Welleck, Amir Yazdanbakhsh, and Peter Clark. Self-refine: Iterative refinement with self-feedback. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*, 2023.
- [13] Chen Qian, Wei Liu, Hongzhang Liu, Nuo Chen, Yufan Dang, Jiahao Li, Cheng Yang, Weize Chen, Yusheng Su, Xin Cong, Juyuan Xu, Dahai Li, Zhiyuan Liu, and Maosong Sun. Chatdev: Communicative agents for software development. In *Proceedings of the 2024 Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024.
- [14] Noah Shinn, Federico Cassano, Edward Berman, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning. *arXiv preprint arXiv:2303.11366*, 2023.
- [15] Saizhuo Wang, Hang Yuan, Leon Zhou, Lionel M. Ni, Heung-Yeung Shum, and Jian Guo. Alpha-gpt: Human-ai interactive alpha mining for quantitative investment. *arXiv preprint arXiv:2308.00016*, 2023.
- [16] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations (ICLR)*, 2023.

- [17] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *International Conference on Learning Representations (ICLR)*, 2022.
- [18] Qingyun Wu, Gagan Bansal, Jieyu Zhang, Yiran Wu, Beibin Li, Erkang Zhu, Li Jiang, Xiaoyun Zhang, Shaokun Zhang, Jiale Liu, Ahmed Hassan Awadallah, Ryen W. White, Doug Burger, and Chi Wang. Autogen: Enabling next-gen llm applications via multi-agent conversation. *arXiv preprint arXiv:2308.08155*, 2023.
- [19] Shijie Wu, Ozan Irsoy, Steven Lu, Vadim Dabravolski, Mark Dredze, Sebastian Gehrmann, Prabhanjan Kambadur, David Rosenberg, and Gideon Mann. Bloomberggpt: A large language model for finance. *arXiv preprint arXiv:2303.17564*, 2023.
- [20] Hongyang Yang, Xiao-Yang Liu, and Christina Dan Wang. Fingpt: Open-source financial large language models. In *Proceedings of the FinLLM Symposium at IJCAI 2023*, 2023.
- [21] Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Thomas L. Griffiths, Yuan Cao, and Karthik Narasimhan. Tree of thoughts: Deliberate problem solving with large language models. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*, 2023.
- [22] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. React: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations (ICLR)*, 2023.
- [23] Hang Yuan, Saizhuo Wang, and Jian Guo. Alpha-gpt 2.0: Human-in-the-loop ai for quantitative investment. *arXiv preprint arXiv:2402.09746*, 2024.